

Databases and Database Users

Databases and database systems are an essential component of life in modern society: most of us encounter several activities every day that involve some interaction with a database. For example, if we go to the bank to deposit or withdraw funds, if we make a hotel or airline reservation, if we access a computerized library catalog to search for a bibliographic item, or if we purchase something online—such as a book, toy, or computer—chances are that our activities will involve someone or some computer program accessing a database. Even purchasing items at a supermarket often automatically updates the database that holds the inventory of grocery items.

These interactions are examples of what we may call **traditional database applications**, in which most of the information that is stored and accessed is either textual or numeric. In the past few years, advances in technology have led to exciting new applications of database systems. The proliferation of social media Web sites, such as Facebook, Twitter, and Flickr, among many others, has required the creation of huge databases that store nontraditional data, such as posts, tweets, images, and video clips. New types of database systems, often referred to as **big data** storage systems, or **NOSQL systems**, have been created to manage data for social media applications. These types of systems are also used by companies such as Google, Amazon, and Yahoo, to manage the data required in their Web search engines, as well as to provide **cloud storage**, whereby users are provided with storage capabilities on the Web for managing all types of data including documents, programs, images, videos and emails. We will give an overview of these new types of database systems in Chapter 24.

We now mention some other applications of databases. The wide availability of photo and video technology on cellphones and other devices has made it possible to

store images, audio clips, and video streams digitally. These types of files are becoming an important component of **multimedia databases**. **Geographic information systems (GISs)** can store and analyze maps, weather data, and satellite images. **Data warehouses** and **online analytical processing (OLAP)** systems are used in many companies to extract and analyze useful business information from very large databases to support decision making. **Real-time** and **active database technology** is used to control industrial and manufacturing processes. And database **search techniques** are being applied to the World Wide Web to improve the search for information that is needed by users browsing the Internet.

To understand the fundamentals of database technology, however, we must start from the basics of traditional database applications. In Section 1.1 we start by defining a database, and then we explain other basic terms. In Section 1.2, we provide a simple UNIVERSITY database example to illustrate our discussion. Section 1.3 describes some of the main characteristics of database systems, and Sections 1.4 and 1.5 categorize the types of personnel whose jobs involve using and interacting with database systems. Sections 1.6, 1.7, and 1.8 offer a more thorough discussion of the various capabilities provided by database systems and discuss some typical database applications. Section 1.9 summarizes the chapter.

The reader who desires a quick introduction to database systems can study Sections 1.1 through 1.5, then skip or browse through Sections 1.6 through 1.8 and go on to Chapter 2.

1.1 Introduction

Databases and database technology have had a major impact on the growing use of computers. It is fair to say that databases play a critical role in almost all areas where computers are used, including business, electronic commerce, social media, engineering, medicine, genetics, law, education, and library science. The word *database* is so commonly used that we must begin by defining what a database is. Our initial definition is quite general.

A **database** is a collection of related data.¹ By **data**, we mean known facts that can be recorded and that have implicit meaning. For example, consider the names, telephone numbers, and addresses of the people you know. Nowadays, this data is typically stored in mobile phones, which have their own simple database software. This data can also be recorded in an indexed address book or stored on a hard drive, using a personal computer and software such as Microsoft Access or Excel. This collection of related data with an implicit meaning is a database.

The preceding definition of database is quite general; for example, we may consider the collection of words that make up this page of text to be related data and hence to

¹We will use the word *data* as both singular and plural, as is common in database literature; the context will determine whether it is singular or plural. In standard English, *data* is used for plural and *datum* for singular.

constitute a database. However, the common use of the term *database* is usually more restricted. A database has the following implicit properties:

- A database represents some aspect of the real world, sometimes called the **miniworld** or the **universe of discourse (UoD)**. Changes to the miniworld are reflected in the database.
- A database is a logically coherent collection of data with some inherent meaning. A random assortment of data cannot correctly be referred to as a database.
- A database is designed, built, and populated with data for a specific purpose. It has an intended group of users and some preconceived applications in which these users are interested.

In other words, a database has some source from which data is derived, some degree of interaction with events in the real world, and an audience that is actively interested in its contents. The end users of a database may perform business transactions (for example, a customer buys a camera) or events may happen (for example, an employee has a baby) that cause the information in the database to change. In order for a database to be accurate and reliable at all times, it must be a true reflection of the miniworld that it represents; therefore, changes must be reflected in the database as soon as possible.

A database can be of any size and complexity. For example, the list of names and addresses referred to earlier may consist of only a few hundred records, each with a simple structure. On the other hand, the computerized catalog of a large library may contain half a million entries organized under different categories—by primary author's last name, by subject, by book title—with each category organized alphabetically. A database of even greater size and complexity would be maintained by a social media company such as Facebook, which has more than a billion users. The database has to maintain information on which users are related to one another as *friends*, the postings of each user, which users are allowed to see each posting, and a vast amount of other types of information needed for the correct operation of their Web site. For such Web sites, a large number of databases are needed to keep track of the constantly changing information required by the social media Web site.

An example of a large commercial database is Amazon.com. It contains data for over 60 million active users, and millions of books, CDs, videos, DVDs, games, electronics, apparel, and other items. The database occupies over 42 terabytes (a terabyte is 10^{12} bytes worth of storage) and is stored on hundreds of computers (called servers). Millions of visitors access Amazon.com each day and use the database to make purchases. The database is continually updated as new books and other items are added to the inventory, and stock quantities are updated as purchases are transacted.

A database may be generated and maintained manually or it may be computerized. For example, a library card catalog is a database that may be created and maintained manually. A computerized database may be created and maintained either by a group of application programs written specifically for that task or by a

database management system. Of course, we are only concerned with computerized databases in this text.

A **database management system (DBMS)** is a computerized system that enables users to create and maintain a database. The DBMS is a *general-purpose software system* that facilitates the processes of *defining*, *constructing*, *manipulating*, and *sharing* databases among various users and applications. **Defining** a database involves specifying the data types, structures, and constraints of the data to be stored in the database. The database definition or descriptive information is also stored by the DBMS in the form of a database catalog or dictionary; it is called **meta-data**. **Constructing** the database is the process of storing the data on some storage medium that is controlled by the DBMS. **Manipulating** a database includes functions such as querying the database to retrieve specific data, updating the database to reflect changes in the miniworld, and generating reports from the data. **Sharing** a database allows multiple users and programs to access the database simultaneously.

An **application program** accesses the database by sending queries or requests for data to the DBMS. A **query**² typically causes some data to be retrieved; a **transaction** may cause some data to be read and some data to be written into the database.

Other important functions provided by the DBMS include *protecting* the database and *maintaining* it over a long period of time. **Protection** includes *system protection* against hardware or software malfunction (or crashes) and *security protection* against unauthorized or malicious access. A typical large database may have a life cycle of many years, so the DBMS must be able to **maintain** the database system by allowing the system to evolve as requirements change over time.

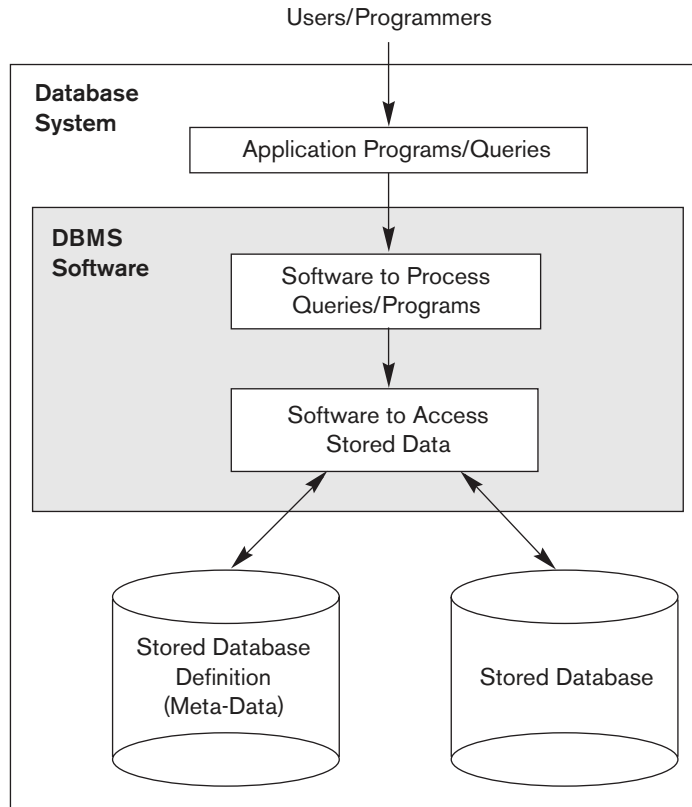
It is not absolutely necessary to use general-purpose DBMS software to implement a computerized database. It is possible to write a customized set of programs to create and maintain the database, in effect creating a *special-purpose* DBMS software for a specific application, such as airlines reservations. In either case—whether we use a general-purpose DBMS or not—a considerable amount of complex software is deployed. In fact, most DBMSs are very complex software systems.

To complete our initial definitions, we will call the database and DBMS software together a **database system**. Figure 1.1 illustrates some of the concepts we have discussed so far.

1.2 An Example

Let us consider a simple example that most readers may be familiar with: a UNIVERSITY database for maintaining information concerning students, courses, and grades in a university environment. Figure 1.2 shows the database structure and a few sample data records. The database is organized as five files, each of which

²The term *query*, originally meaning a question or an inquiry, is sometimes loosely used for all types of interactions with databases, including modifying the data.

**Figure 1.1**

A simplified database system environment.

stores **data records** of the same type.³ The STUDENT file stores data on each student, the COURSE file stores data on each course, the SECTION file stores data on each section of a course, the GRADE_REPORT file stores the grades that students receive in the various sections they have completed, and the PREREQUISITE file stores the prerequisites of each course.

To *define* this database, we must specify the structure of the records of each file by specifying the different types of **data elements** to be stored in each record. In Figure 1.2, each STUDENT record includes data to represent the student's Name, Student_number, Class (such as freshman or '1', sophomore or '2', and so forth), and Major (such as mathematics or 'MATH' and computer science or 'CS'); each COURSE record includes data to represent the Course_name, Course_number, Credit_hours, and Department (the department that offers the course), and so on. We must also specify a **data type** for each data element within a record. For example, we can specify that Name of STUDENT is a string of alphabetic characters, Student_number of STUDENT is an integer, and Grade of GRADE_REPORT is a

³We use the term *file* informally here. At a conceptual level, a *file* is a *collection* of records that may or may not be ordered.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2

A database that stores student and course information.

single character from the set {'A', 'B', 'C', 'D', 'F', 'I'}. We may also use a coding scheme to represent the values of a data item. For example, in Figure 1.2 we represent the Class of a STUDENT as 1 for freshman, 2 for sophomore, 3 for junior, 4 for senior, and 5 for graduate student.

To *construct* the UNIVERSITY database, we store data to represent each student, course, section, grade report, and prerequisite as a record in the appropriate file. Notice that records in the various files may be related. For example, the record for Smith in the STUDENT file is related to two records in the GRADE_REPORT file that specify Smith's grades in two sections. Similarly, each record in the PREREQUISITE file relates two course records: one representing the course and the other representing the prerequisite. Most medium-size and large databases include many types of records and have *many relationships* among the records.

Database *manipulation* involves querying and updating. Examples of queries are as follows:

- Retrieve the transcript—a list of all courses and grades—of 'Smith'
- List the names of students who took the section of the 'Database' course offered in fall 2008 and their grades in that section
- List the prerequisites of the 'Database' course

Examples of updates include the following:

- Change the class of 'Smith' to sophomore
- Create a new section for the 'Database' course for this semester
- Enter a grade of 'A' for 'Smith' in the 'Database' section of last semester

These informal queries and updates must be specified precisely in the query language of the DBMS before they can be processed.

At this stage, it is useful to describe the database as part of a larger undertaking known as an information system within an organization. The Information Technology (IT) department within an organization designs and maintains an information system consisting of various computers, storage systems, application software, and databases. Design of a new application for an existing database or design of a brand new database starts off with a phase called **requirements specification and analysis**. These requirements are documented in detail and transformed into a **conceptual design** that can be represented and manipulated using some computerized tools so that it can be easily maintained, modified, and transformed into a database implementation. (We will introduce a model called the Entity-Relationship model in Chapter 3 that is used for this purpose.) The design is then translated to a **logical design** that can be expressed in a data model implemented in a commercial DBMS. (Various types of DBMSs are discussed throughout the text, with an emphasis on relational DBMSs in Chapters 5 through 9.)

The final stage is **physical design**, during which further specifications are provided for storing and accessing the database. The database design is implemented, populated with actual data, and continuously maintained to reflect the state of the miniworld.